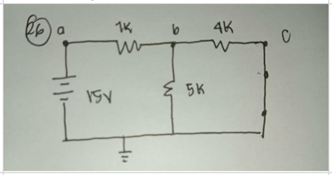
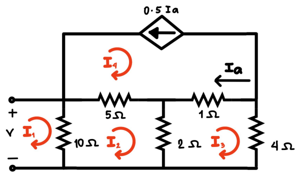
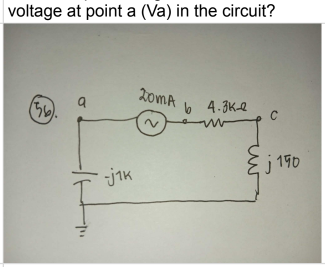
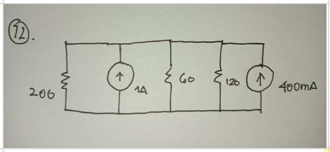
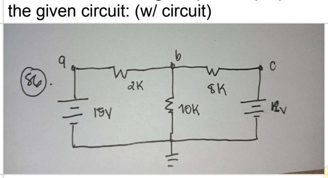
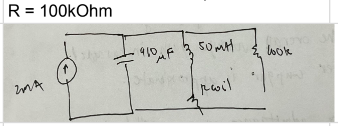
ELECS	QUESTION	A	B	C	D
1	If a 4 bit Johnson counter has an initial state of 1001. What is the state after the second clock pulse?	0110	0000	1111	1010
2	What factor significantly contributes to the high-frequency response of a BJT amplifier?	The emitter resistance	The Collector-Emitter Capacitance	The Base-Emitter junction capacitance	The power supply filtering
3	How is the impulse response of an LTI system is defined?	The system's response to a Dirac delta function input	The system's response to a unit step input	The system's response to a sinusoidal input	The system's response to a ramp input
4	When connecting two amplifiers in a cascade, what should be the value of the second amplifier's input impedance in relation to the first amplifier's output impedance?	Equal to the first stage output impedance	The same as the load impedance	Low, to match the first stage's output impedance	High, to minimize loading on the first stage
5	A multiplexer with 5 control signals can control how many inputs?	32		16	25
6	Three resistors of 25 ohms, 10 ohms, and 50 ohms are connected in series with a power source of 25 Volts. What is the voltage drop at the second resistor?	2.94V	18.14 V	7.35 V	15.70 V
7	This tool is required to consolidate its various modules into a single executable file. What tool should be used to address this requirement?	Assembler	Loader	Compiler	Linker
8	In a series RC circuit where Vc = 15 V and Vr = 20 V, how much is the total voltage?	35V	65 V	15 V	25 V
9	What type of electroacoustic transducer converts the electrical charges produced by some forms of solid materials into energy?	Chemical	Piezoelectric	Mutual Induction	Passive
10	In the transfer function G(s) = 10 / (s^2 + 5s + 6), the system is classified as:	First Order	Third Order	Zero Order	Second Order
11	In a magnetically coupled circuit, how does increasing air gap in the core affect the mutual inductance?	It decreases the mutual inductance	It increases mutual inductance	It does not affect mutual inductance	It decreases the self inductance but not the mutual inductance
12	Which statement is TRUE about Zener diodes?	Prevent current from exceeding a certain level	Lowers forward voltage drop than standard diodes	maintains a stable voltage across during operation	Operates in the reverse breakdown voltage
13	How can this be mitigated?	Increase the gate-source capacitance Cgs	Increases the transconductance	Reduce the gain of the amplifier	Increase the drain resistance RD
14	A Building Management System (BMS) is implemented in the building using technologies like AI, IoT, and data analysis. The property administrator is paying a lot to Meralco due to multiple equipment in operation. How can energy optimization be achieved by the Building Management System (BMS)?	Montoring and controlling key building systems	Scheduling of Building Operation	Predicting fault and alarm occurrence	Understanding energy demands and assessing them
15	What is the source voltage of the circuit?	128V	438	328 V	158 V
16	Which power supply compoent is responsible for converting AC voltage to pulsating DC voltage?	Regulator	Rectifier	Voltage multiplier	Filter Capacitor
17	Which JFET amplifier has a high Z_in, a low Z_out, and a voltage gain less than one?	Common-source amplifier	Common-emitter	Common-gate amplifier	Source follower
18	A feedback control system has a forward path gain of 10 and a feedback path gain of 2. What is the resulting primary feedback ratio of this system?	2/21	10/21	20/21	1/21
19	How many flip-flops are needed to represent all possible states?	4 flip-flops	6 flip-flops	5 flip-flops	7 flip-flops
20	Your team is creating an ASM chart for a traffic light controller. Curlee says you place states like Red Light and Green Light inside an oval. Julius uses diamond to indicate conditions such as "Is Timer Expired?" Sarah places the action "turn on yellow light" inside a rectangle. Who among the three followed the correct ASM chart conversion?	Sarah only	Curlee Only	Julius only	Julius and Sarah
21	What is the maximum stored energy Wmax?	37.5 mJ	150.5 mJ	1.87 mJ	75.0 mJ
22	A battery can supply 16 J of energy to move 10 C of charge. What is the voltage of the battery?	160V	0.625 V	4.33V	1.6V
23	Which is the best approach to do an integrated solution?	Disconnecting the fire alarm from the access control system to avoid interference	Manually unlocking all doors upon firealarm activation	Making the security doors remain unlocked during time of emergencies	Using a relay-based trigger to unlock doors during fire alarms
24	The simplified Boolean expression for f(A, B,C) = A · (B + C) + A · (B' + C') + A · B is ?	f (A,B,C) = A	f (A,B,C) = A•B•C	f (A,B,C) = A•B•C	f(A, B, C) = A•B+C
25	Find the collector voltage Vc?	4.5 V	5.3 V	3.5 V	8 V

26	Using the superposition principle of the circuit with a source voltage of 15 V, what is the voltage at point b (Vb)?	1.034V	10.34 V	8.034V	12.34 V
					
27	What will be the output voltage?	1.67 V	2.50 V	2.67 V	1.50 V
28	A mod-8 counter requires how many flip-flops?	3	2	4	8
29	Since thousand of sine waves from different radio stations hit the antenna, what will be the role of the oscillator?	Tuner	Feedback network	Controller	Amplifier
30	In practical application, what system instructs or tells the computer what to do when it boots up?	Read-Only Memory	Basic I/O System	Embedded Systems Memory	Random Access Memory
31	What is the most effective way to prevent unauthorized access due to credential theft?	Apply multi-factor authentication with biometrics	Use of centralized database for access logs	Use of RFID card readers with PIN entry	Use CCTV to catch the theft
32	A differential amplifier implemented with two bipolar junction transistors has the following parameters, V_CC = - V_EE = 15V; R_C = R_E = 10 kOhms; β_1 = β_2 = 250; R_B1 = R_B2 = 1kOhms. What will be the resulting tail current for this circuit?	71.5 μA	1.43 mA	2.25 mA	2.86 μA
33	Which circuit modification could be the most effective in preventing this incident?	Adding a series capacitor across the SCR's anode and cathode	Placing a resistor parallel with the gate	Using snubber circuit (a resistor and capacitor in series) across the SCR	Connecting an additional diode in reverse bias across the gate
34	In a root locus plot, what is the result of adding a pole to the system?	Shifting the poles further into the left hand plane	Increasing the phase Margin	Moving the root locus closer to the imaginary axis	Decreasing the system gain
35	Race condition in asynchronous sequential circuit?	The state machine's clock is not synchronized	a Feedback loop causes an unstable output	The circuit cannot decide on the next state	The outputs stabilize after a long day
36	A microprocessor with a 20-bit address bus is connected to memory, what is the maximum addressable memory size?	256kB	2.5 MB	1.0MB	512 kB
37	What will happen to the system if the system's phase margin is negative?	Marginally stable	Unstable	Oscillatory	Unstable
38	The input frequency is 1 kHz connected in a series circuit with a capacitive reactance of -j200; inductive reactance of j600, and a 750 ohms resistor. What is the capacitor value?	586 nF	200 nF	986 nF	796 nF
39	power receive per unit area?	Irradiance	Luminous Intensity	Luminance	Radiant intensity
40	You are designing a digital circuit what needs to output HIGH only when three specific conditions are true simultaneously. The logic must be implemented using ICs from 74xx family. Which IC would be the MOST efficient choice to minimize the number of gates and wiring?	7400	7402	7411	7432
41	Vs = 1 Vp, R = 140 Ohms, C = 50 nF, and L = 22 mH. What is resonant frequency?	1.06 kHz	1 kHz	4.80 kHz	2.80 kHz
42	Calculate the floating-point rate for a single processor given the data. The HP 9000 Superdome has two operations per cycple times the floating-point units per processor yields a rate of 4 operations per cycle per processor. The processor frequency (clockspeed) is F = 1.1 GHz it has a processor cycle time (1/F) of t = 909 ps, and the floating-point operations.	0.1690 TF	0.0044 TF	0.00528 TF	0.00492 TF
43	Which is true about PIN diode?	Low reactance in the forward direction	High impedance in reverse direction	High reactance in the reverse direction	High impedance in the forward direction
44	A system is experiencing a potential interrupt conflict. How would you use the Read, Interrupt, Mask (RIM) instruction to determine the status of the program?	Interpret the accumulator bits	Execute interrupt mark status	Load the bits simulator	Assess possible conflicts
45	In the small signal model of diode. What component PRIMARILY represent the diode's resistance?	Breakdown Resistance (Rb)	Series-resistance	Shunt Capacitance (Cj)	Dynamic Resistance (rd)
46	What is the utility program that translates assembly language source code into executable machine code?	Compiler	Assembler	Register	Instruction
47	What happens to the non-linear distortion of the output signal caused by the initiation of the negative feedback?	Decreases	Increases	Changes sinusoidal	remains stable
48	What is the circuit component PRIMARILY contributes to the high frequency gain loss of a BJT amplifier?	The supply ripple	The coupling capacitor	The Base-Emitter junction capacitance	The emitter resistor

49	Which condition results in the HIGHEST voltage gain (Av) in an amplifier?	Load resistance at its minimum value with a high source resistance	Load resistance at its maximum value with a low source resistance	Load resistance equal to the source resistance with both at medium values	Load resistance at its maximum value with a low source resistance
50	Given the following conditions, what is the resulting input impedance of the network?	3.517	3.911 ohms	3.711 ohms	3.458 ohms
					
51	What determines the frequency response of an op-amp?	The input impedance and overall frequency	The open loop gain and the bandwidth product	The output impedance and overall frequency	The power supply voltages and bandwidth product
52	What type of circuit is used for overvoltage protection?	Thermoster	Crowbar	Suppressors	Clipper
53	Technician Roi says a relay is used to create a voltage drop in a control circuit. Whose statement is TRUE about relay?	Technician Roi	Neither Technician Paul nor Roi	Both Technician Paul and Roi	Technician Paul
54	In a 3-input OR gate, what will be the output when all inputs are 0?	Undefined	1	0	Infinity
55	The Boolean function f(A, B, C) = A(B+C) + A'(B'+C) can be reduced to:	f(A,B,C) = A · C + B	f(A,B,C)=AB+A'B'+C	f(A,B,C) = A + C	f(A, B, C) = A · B' + A · B' C' + C'
56	Assume that the current source is the reference phase angle of 0°. What is the voltage at point a (Va) in the circuit?	-25.5∠ -2° V	-20.9 ∠ -90° V	-45<-90° V	-86.05 ∠ -2° V
					
57	What is measured by mutual inductance between two coils?	Total flux lnking the coils	Voltage induced in one coil by a current in the other	Inductance of the individual coils	Resistance of the coils
58	The Boolean function f(A,B,C) = A' · (B+C) + A' · (B'+C) can be simplified to:	f (A,B,C) = A +C	f (A,B,C) = A•B + C	f (A,B,C) = A'	f(A,B,C) = A + B + C
59	What step should the engineer follow to provide a microcontroller with the I/O signals to the device?	Use dedicated I/O ports and peripherals	Convert analog to digital data for device processing	Use of communication ports to devices	Send digital pulses to external devices
60	What is the purpose of a coupling capacitor?	To block high-frequency noise	To follow AC signals to pass while blocking DC components	To block the low frequency noise	To allow DC signals to pass while blocking AC components
61	What values of "S" make the denominator equal to 0, defining the poles of the transfer function?	Negative	One	Zero	Positive
62	A BJT small-signal amplifier has a capacitor bypassing the emitter resistor. What happens if the capacitor is removed?	The DC operating point changes significantly	The frequency response shifts to lower frequency	The gain of amplifier decreases	The input impedance dereases
63	However, the measured output is only 60V. What is the most probable cause of this result?	The capacitors are too small, limiting charge storage	The input frequency is too high, preventing full capacitor charging	The diodes have excessive reverse leakage current	The load resistance is too high, causing voltage collapse
64	The characteristic equation of a given system is $s^4 + 6s^3 + 11s^2 + 6s + k = 0$. What restrictions must be placed upon the parameter k to ensure that the system is stable?	-1<K<5	0<K<15	0<K<10	-3 < K < 3
65	What is the corresponding frequency?	250 Hz	2kHz	32 kHz	4 kHz
66	What is the main PURPOSE of using feedback in an amplifier circuit?	To increase amplifier gain and reduce distortion	To increase the output impedance and reduce distortion	To improve bandwith and reduce distortion	To increase the input resistance and reduce distortion
67	The overall voltage gain in a cascaded amplifier configuration is approximately equal to the:	Sum of the individual voltage gains	Difference of the individual voltage gains	Average of the individual voltage gains	Product of the individual voltage gains
68	What does Gauss's Law for magnetism state in Maxwell's equations?	The net magnetic charge in a region is zero	The magnetic field is due to electric current only	The magnetic flux density through a surface is proportional to the enclosed magnetic charge	The electric flux through a surface is proportional to the enclosed charge
69	Admittance of parallel circuit is 0.12 ∠−30° S. The circuit is:	Inductive	Reactive	Resistive	Capacitive
70	What theorem states that a circuit with multiple power sources can be analyzed by evaluating only one power source at a time?	A Node voltage method	Thevenin's	Maximum Power Transfer	Superposition

71	When an electronic circuit replicates a current from one branch tro another and acts like current amplifier, this is called:	Differential Circuit	Source Voltage	Current Source	Current Mirrors
72	A parallel netowrk with two current sources. Determine the current through the 200 ohms resistor.	3.42 mA	243 mA	123 mA	233 mA
					
73	In a common-emitter amplifier, what condition must be met to ensure a high voltage gain?	Reduce emitter resistance while increasing collector resistance	Reduce collector resistance while keeping emitter resistance constant	Increase base resistance while reducing collector resistance	Increase emitter resistance while keeping collector resistance constant
74	Two inductances L_1 = 10 mH and L_2 = 20 mH are connected in parallel circuit. What is the equivalent inductance?	12.12mH	7.66 mH	4.37 mH	6.67 mH
75	BJT for high frequency operation, the overall gain-bandwidth product is PRIMARILY influenced by what factor?	The collector-emitter capacitance	The biasing network	The input and output coupling capacitors	The emitter resistance
76	A certain 120 V_AC RMS 60 Hz 1.2 HP motor has an efficiency of 90% and a lagging power factor of 0.85. What reactive power is drawn from the system? (Assume: 1 HP ≈ 745.7 Watts)	817 VAR	975 VAR	617 VAR	770 VAR
77	The two port system Z = [10^3 10; 10^6 10^4] is driven by an ideal sinusoidal voltage source in series with a 500Ω resistor and terminated in a 10kΩ load resistor. What is the voltage gain?	-555	-450	-580	-333
78	In a D-flip flop under what condition will the output Q follow the input D?	The clock is low	The clock is high	The clock edge triggers the flip-flop	The reset signal is active
79	How does the magnetic field (inside the conductor?) behave in a cylindrical conductor carrying a steady current?	An exponential decay	A linear relationship with radius	A constant value	An inverse square law
80	Under what condition will a P-channel JFET operate that could allow current to flow from the source?	Biased positively	Connected to the ground	Biased negatively	Zero Voltage
81	What type of circuit transforms current and creates opposition on the process of current flow?	Lumped	Tuned	Inductive	Linear
82	What type of memory has the advantage of being rewritable but needs to be physically removed from a circuit to be erased and reprogrammed?	Static RAM	Flash eRAM	Register	Electrically Programmable ROM (EPROM)
83	What describes the Curie temperature of a ferromagnetic material?	It losses is ferromagnetic properties and becomes paramagnetic	The permeability becomes infinite	It becomes a perfect conductor	It achieves maximum magnetization
84	What industrial computer control system monitors the state of input and output devices based on a custom program?	Distributed Control Systems	Programmable Logic Controller (PLC)	Supervisory Control and Data Acquisition	Remote Terminal Units
85	When will the output of 3-input AND gate become 1?	At least one input is one	At least one input is 0	Any two inputs are 1	All inputs are 1
86	Determine the voltage at point b (Vb) in the given circuit: (w/ circuit)	16.5V	12.41 V	21.0 V	24.38 V
					
87	2mA current source give the circuit with Rcoil = 100 Ohm, C = 910pF, L = 50mH, R = 100kOhm	f1 = 23.47 kHz and f2 = 25.84 kHz	f1=22.07 kHz and f2=24.15 kHz	f1 = 23.44 kHz and f2 = 23.76 kHz	f1 = 22.56 kHz and f2 = 24.64 kHz
					
88	What part of a computer system manages the storage and retrieval of data and instructions for the processor?	I/O Subsytem	User Interface	Memory Subsystem	Central Processing Unit
89	What is the PURPOSE of the source code digitalWRITE(pin, value) in an Arduino?	Reads the value from a specfied digital PIN, either HIGH or LOW, and returns this value	It can read or write signals that are HIGH or LOW only	Configures the specified pin to behave either as an I/O with HIGH or LOW	Writes a HIGH or a LOW value to a digital pin
90	What is the use of summing point in a block diagram representation of control system?	Multiply Signal	Integrate the signals	Amplify the signals	Add or Subtract Signals
91	.... in the t-parameters for the cascaded networks as shown in the circuit diagram?	3.32	4.22	25.84	0.19

92	What will happen to a charged particle that enters a uniform magnetic field perpendicularly?	moves in a straight line	Accelerate in the field direction	Follows a helical path	Moves in a circular path
93	Which oscillator uses a positive feedback loop for sustained oscillations?	Low-pass filter	Differential amplifier	Voltage-Controlled Oscillator	Hartley Oscillator
94	An inverting amplifier gain is sensitive to the Miller Effect, what will happen to the input impedance of the amplifier due to the influence of parasitic capacitance?	Decreased	Unaffected	Increased	Phase Change
95	Which is NOT relevant to the functioning of the optoelectronic devices?	Light Control	Light detection	Light Generation	Light Simulation
96	A 47 μF capacitor is connected in series with a 5 kOhms resistor and a 24 V DC supply. What is the initial rate of voltage rise across the capacitor immediately after the true DC supply is connected?	88 V/s	102 V/S	67 V/S	138 V/s
97	Convert 110011110101_2 to octal	1743	3317	6365	6535
98	Three resistors of 25 ohms, 50 ohms, and 100 ohms are connected in parallel with a power source of 100 volts. What is the voltage drop at the second resistor?	14.28	28.57	57.14	100
99	What component of the Ampere's Law contributes to the displacement current?	The time-varying electric field in a dielectric	Current flows due to Magnetic Field	The electric field that induces magnetic field in free space	The induced voltage due to the magnetic flux
100	A parallel AC circuit that applied 120 V_ac resulted to a total current of 5 A. If the phase angle of the circuit is -53.13 degrees, how much real power is dissipated by the circuit?	260 W	480 W	600 VA	360 W

MATH	**QUESTION**	**A**	**B**	**C**	**D**
1	Which condition satisfied for a function f (X) to be continous at x = 0?	f(c) exists	$\lim_{(x\to c)}$ f(x) exists	f(c) = lim (x → c) f(x)	All of the above
2	The power dissipation in an amplifier circuit depends on the input voltage V and load resistance R. Given P(V,R) = 2V^2/R + 0.1R, what is the rate of change or power with respect to voltage when V=5 and R=10?	1.0 W/V	4.0 W/V	3.0 W/V	2.0 W/V
3	If f''(X) > 0 what can be concluded?	The function is concave up	The function is concave down	The function has a maximum	The function is decreasing
4	A robotic arm moves according to x(t) = 3t^2 - 2t and y(t) = 4t + 1. What is the speed of the arm at t = 1?	√31 units	6 units	2√13 unit/s	4√2 unit/s
5	What path does the electron trace?	Straight line	Parabola	Circle	Ellipse
6	What is lim x→∞ $(2x^3 - 5x^2 + 1) / (x^3 + 3x - 7)$?	0	1	2	∞
7	Find the equation of the tangent line to curve x = t^3 - t, y = t^2+2 at t=1	I. y=(2/3)x + 7/3	IV	III. V = (1/2)x + 5/2	II. y = x + 3
8	The efficiency η of a transformer is given by η(x) = 100x/(x + 2 + 50/x), where x is the load current in amperes. What load current maximizes efficiency?	5.0A	4.03 A	10.0 A	7.07 A
9	Find the x component of the electric field Ex = -dV/dx at point (1,2)?	4 V/m	2 V/m	-4 V/m	-2 V/m
10	Find the absolute minimum value of f(x) = $x^4 - 8x^2 + 5$ on the interval [-2, 3].	-11	-9	5	14
11	What is the total time of flight before it returns to ground? (g=9.81)	2.65 s	3.06s	4.45s	5.30 s
12	For the function g(x) = e^{2x} sin(x), what is g''(0)?	0	2	4	5
13	Find dy/dx at point (1,1) for curve x^2y + xy^2=2	-1	-2/3	-3/2	1
14	What is lim x→0 x^2 / |x|?	0	1	-1	Does not exist
15	What does the second derivative x''(t) physicaly represent?	Velocity of the mass	Dampnig force on the mass	Spring constant	Acceleration of the mass
16	Why is the integral ∫(from 0 to 1) 1/√x dx considered improper?	The integrand is negative for some X	The integral automatically converges	The upper limit is finite	the integrand is unbounded at x = 0
17	What is the formula for the volume of the solid revolution rotating x-axis from a to b ?	V=∫ f(x) dx	V = π ∫ f(x) dx	v = π ∫ [f(x)]^2 dx	V = 2π ∫ x f(x) dx
18	Evaluate the double integral ∬_R xy dA where R is the rectangle 0 ≤ x ≤ 2, 0 ≤ y ≤ 3	6	18	12	9
19	What makes the integral 1/x^2 from -1 to 1 improper?	The limits are symmetric	The upper limit is finite	The integrand is always positive	The integrand has a discontinuity at x = 0
20	A signal decays according to f(t) = 6e^(-0.5t). What is the total area under the curve from t =0 to t = ∞?	infinity	18	6	12
21	What is the most appropriate technique for ∫ e^ax dx	Integration by parts	Trigonometric Substitution	Algebraic substitution	Partial fraction decomposition
22	If ρ(x,y) represents the density at point (x, y) in a region R, what does ∬_R ρ(x,y) dA represent?	The volume above region R under the curve	The total mass of the region	The area of region R	the centroid of R
23	What is the formula of ∫ e^ax dx where a is a constant?	e^ax + C	ae^(ax) + C	(1/a) e^(ax) + C	$(1/a^2)$ e^(ax) + C
24	Which integral requires trigonometric substitution to solve?	III.	IV. integral x / (x^2 + 1) dx	I. (∫(dx/x^2+4))	II.∫ $\sqrt{a - x^2}$ dx
25	What is the displacement from t=0 to t-s seconds?	8m	6m	5m	4 m
26	If F(x) = ∫(from a to x) f(t) dt, what does F'(x) equal?	f(a)	f(t)	F(x)	f(x)
27	Evaluate ∫ 6x^2 cos(x^3) dx usng substitution.	2sin(x) + C	$\sin(x^3)$ + c	2 $\cos(x^3)$ + C	−2 $\cos(x^3)$ + C
28	Which method uses cylindrical shells to find the volume of revolution?	Disk Method	Washer method	Shell method	Cross sectional area method
29	Wht is the averag evalue of f(x) = x^2 on the interval [0,3]?	3	6	9	27
30	What integral represents the arc length of y = √x from x = 1 to x = 4?	I. $\int_1^4 \sqrt{1 + 1/4x}\,dx$)	II. $\int_1^4 \sqrt{1 + 1/x}\,dx$)	III. ($\int_1^4 \sqrt{1 + x}\,dx$)	IV. $\int_1^4 \sqrt{x + 1}\,dx$
31	Differential equaion v(t) across an inductor in RC with R and C and v(t)?	I. L(di/dt) + Ri=V(t)	II	III. RC(dv/dt)+ v = V(t)	IV. LC (d^2i/dt^2) + i = V(t)
32	A 10 uF capacitor charged to 12 V is discharged through a 100-kOhm resistor. What is the voltage across the capacitor after 2 seconds?	4.12V	7.22 V	6.25 V	1.63 V
33	What is the genral solution of dy/dx = 2xy?	y = Ce^{x^2}	y = C e^{2x}	y=x^2 + C	y = Cx^2
34	Which first order ODE models the charge q(t) on a capacitor in an RC circuit with a constant current source I_o?	III. c(dq/dt) =Io	IV. L(dq/dt) + Rq = Io	I. (RC(dq/dt) + q/C = I_oR)	II.

35	A step voltage of 50 V is applied at t=0. What is the current at t = 1 seconds?	3.16 A	0.86 A	5.00 A	4.32 A
36	A series RLC circuit with R = 6 ohms, L = 1 H, C = 0.04 F has the governing equation L (d^2q/dt^2) + R (dq/dt) + q/C = 0. Given q(0) = 0.1C and q'(0) = 0, which form represents the change q(t)?	I. q(t)=e^{-3t}(Acos(4t)+Bsin(4t)	II. q(t) = Ae^(-2t) + Be^(-4t)	III. (q(t) = (A + Bt)e^{-3t})	IV. q(t) = A cos(5t) + B sin(5t)
37	If 80 mg is administered how much remain after 18 hours?	10 mg	5 mg	20 mg	2.5 mg
38	A spring-mass-damper system obeys the equation 2 (d^2y/dt^2) + 8 (dy/dt) + 8y = 0. What is the nature of the response?	Undamped	Overdamped	Underdamped	Critically Damped
39	Determine the displacement x(t) for t > 0?	x(t) = 3e^{-2t}	x(t) = (3 + 3t)e^{-2t}	x(t)= 3e^{-2t} - 6te^{-2t}	x(t) = 3e^{-2t}(1+t)
40	Findthe inverse Laplace transform of F(s) = (2s + 5)/[(s + 1)(s + 3)].	2e^{-t} + e^{-3t}	(3/2)e^t + 1/2e^(-3t)	e^{-t} + e^{-3t}	(1/2)e^{-t} + (3/2)e^{-3t}
41	In first order RL circuit what does the time constant L/R physically represent?	The time for current to reach maximum value for the first time	The time for current to reach 63.2% of its final value	The time for current to decay to zero	The resonant frequency of the circuit
42	What is the general solution of y''' - 6y'' + 11y' - 6y = 0?	y = C_1e^x + C_2e^2x + C_3e^3x	y = C1e^(-x) + C2e^(-2x) + C3e^(-3x)	y = C_1 + C_2e^x + C_3e^{6x}	y = C_1e^x + C_2xe^x + C_3e^{6x}
43	Using laplace transform what is i(t)?	i(t) = (1/8)(1 − e^{-2t})	i(t) = (1/4)(1 − e^{-2t})	i(t)= (1/2) (1-e^{-2t})	i(t) = 2(1 - e^{-2t})
44	The differential equation RC(dv/dt) + v = Vo describes an RC circuit. What does this equation represent?	The exponential decay of the capacitor voltage from an initial value Vo down to Zero.	The linear increase of voltage across the capacitor determined by the time constant.	The exponential decay of the circuit current as the capacitor reaches full charge	The exponential rise of the capacitor volatage towards the steady - state value Vo
45	What form of solution is assumed when the characteristic equation has complex conjugate roots a+Bi?	y = (C_1 + C_2t)e^{at}	y = $c_1e^{\alpha t}$ + $c_2e^{\beta t}$	y =$e^{\alpha t}$ (C_1cos(βt) + C_2sin(βt))	y = C_1 cos(αt) + C_2 sin(βt)
46	A damped harmonic oscillator is governed by d^2x/dt^2 + 4 dx/dt + 13x = 0 with x(0) = 0 and x'(0) = 9. Which expression correctly represents x(t)?	I. x(t)=3e^{-2t} sin(3t)	II. x(t) = 9e^(-2t) cos(3t)	III. (x(t) = e^{-2t}(3cos(3t) + 9sin (3t))	IV. x(t) = 3e^(-4t) sin(at)
47	What is the approximation after one iteration starting x = 1 to x = 2	1.33	1.5	1.67	1.75
48	For an alternating series Σ $(-1)^n$ a□, which condition guarantees convergence?	I only - an is decreasing and lim (n approches to infinity) an=0	II only. Σ |an| converges	I. or II. (I. A□ is decreasing and lim n -> ∞, II. ∑ |a□| converges)	III. and IV. (I. a□ is bounded, II. an>0 for all n
49	The signal is modeled as x(t) = e^at dt where unit step function and a>0. What is the fourier transform X(W)	X(ω) = 1/(a + jω)	x(ω) = 1/(a − jω)	x(ω) = a/(a^2+ω^2)	x(ω) = jω/(a + jω)
50	Using the vertex formula, whatis the approximate maximum value of x for the funciton f(x) = -2x² + 8x + 3?	1	2	3	4
51	What can be conlcuded about the system?	Unstable due to positive real parts of poles	Stable with oscillatory response	Unstable due to poles on imaginary axis	Marginally stable
52	Which differential equation is linear, homogenous and second-order with constant coefficients?	y''+3y'+2y=e^x	y'' + xy' + 2y = 0	y'' + 4y' - 5y = 0	$(y')^2$ + 2y = 0
53	This representation comes from modelling the signal as sum of the phasor rotating in opposite direction. Which of the following represent?	x(t) = 20sin(2000 + 45°)	x(t) = 20cos(2000t + 45°)	x(t) = 10sin(2000t + 45°)	x(t) = 10 cos(2000t + 45°)
54	What is the MOST likely outcome when solving an overdetermined linear system using least squares method?	Exactly one uniqe solution	Infinitely many solutions	A best fit approximate solution that minimizes error	No solution exist
55	P1 + P2 = 12 and P1^2 + P2^2 = 80 where P1 and P2 are transmitted power in milliwatts of user 1 and user 2. Which of the following represents the possible equilibrium power?	(4,8) and (8,4)	(6,6) and (10,2)	(3,9) and (9,3)	(5, 7) and (7, 5)
56	A silicon diode is connected in series with a 500 ohm resistor and a 10 V DC voltage source. The diode operates at room temperature and has an ideality factor n = 1, a reverse saturated current Is = 5 x10^-12 A, V_t = 0.026 V. The diode obeys the following relation ID = Is(exp(VD/nVT) - 1). Find VD.	VD=0.450V	VD = 0.573V	VD = 0.685 V	VD = 0.820 V
57	If the initial current is i(0) = 0 and the initial derivative is di(0)/dt =8. What is the expression for i(t)?	II. i(t) = 4te^-2t	I	IV. i(t)=8te^{-2t}	III. i(t) = (4 + 4t)e^(-2t)
58	Which problem type specifies conditions at two different points in the domain?	Initial Value problem (IVP)	Boundary value problem (BVP)	Eigenvalue problem	Characteristic Equation
59	Usig trapezoidal rule with 1 interval estimate the total charge delivered	18 C	20 C	14 C	16 C
60	An AC circuit has impedance Z = 6 - 8j ohms. What is the magnitude of the impedance?	2 ohms	10 ohms	14 Ω	100 Ω
61	According to Nyquist Theorem what is the minimum samplig rate required to avoid aliasing?	12 kHz	24 kHz	6kHz	48 kHz
62	In quality control testing of semiconductor devices, what does rejecting Ho indicate?	There is sufficient evidence to support H1	The null hypothesis is definitely false	The sample size was too small	No conclusion can be drawn

63	All of the following are advntges of using fatorial experiment EXCEPT?	Ability to study interaction effects between factors	More efficient use of experimental resources.	Guaranteed elimination of all measurement errors	Simultaneous evaluation of multiple factors
64	Which correlation measure is used when studying the LINEAR relationship between two continuous variables?	spearman's rank coeffcient	Kendall's tau	Pearson correlation coefficient	Point correlation coefficient
65	The distribution is found to be approximate normal with the mean of 10k ohms and standard deviation of 200 ohm. What perentage of resistor are expected to fall within ------------------------	68.20%	95.40%	99.70%	50 %
66	What does the design of an experiment determine regarding variable affecting the state of the process?	The variable that do not effect the process	The variable least influential to the process	The variables that remain constant	The variables most influential to the process
67	To obtain reliable data what is most important consideration during measurement?	Selecting a sampling rate much higher than the frequency content	Using sensors whose frequency responce covers the vibration range of interest	Matching the sampling rate to atleast twice the highest frequency vibration expected	Recording data for a duration long enough to capture the system behavior
68	A semiconductor fabrication process produces chips with an average yield of 85% and standard deviation of 5%. A particular wafer has a yield of 92%. What is the z-score for this wafer?	0.7	1.4	2.1	2.8
69	What does this pattern indicate?	Natural variation in the process	Improvement in process capability	Malfunction of the measurement tool	Potential shift in the process mean
70	A 2x2 factorial experiment tests the effects of bias voltage (A) and frequency (B) on amplifier gain. The following results ware obtained: (table) What is the main effect of factor A?	2db	4dB	6 dB	8dB
71	Which ransformation is most appropriate to achieve approximate normality?	Square transformation	Reciprocal Transformation	logarithmic transformation	No transformation needed
72	Two variables x and y have cov(x,y) = 6, var(x) = 9, and var(y) = 16. What is the correlation coefficient r?	0.25	0.5	0.75	1.00
73	Which sampling strategy ensures each bath is proportional represented?	Simple Random Sampling	Stratified Random Sampling	systematic sampling	Cluster sampling
74	An engineer wants to test whether two PCB soldering techniques (Method A and Method B) affect signal integrity. Since different soldering operations might introduce variability, the operation is treated as a block...***(lost)...*** After analysis, the results showed that the operator (block) effect was significant, while soldering method (treatment) effect was not significant. ***(wasnt able to copy question)***	the presence of block effect indicates that soldering methods interact with operator skills	Blocking failed, since the treatment did not appear significant	The choice of soldering method strongly affects signal integrity, while operator influence is minor	Signal integrity is primarily influenced by the operator's variability not by the soldering method
75	In hypothesis testing what does p-value represent?	The probability that the null hypothesis is false	The confidence level of hypothesis test	The likelihood of collecting new data after the test	The probability of obtaining the observed results assuming the null hypothesis is true
76	Which statement BEST describes the Divergence Theorem?	it converts a line integral into a surface integral	It converts a surface integral into a volume integral	It converts a volume integral into a surface integral	It converts a double integral into a line integral
77	In electromagnetism Faraday Law relate the curl of E?	Time rate of change B	Electric Charge Density	Magnetic Flux Density	Electric potential
78	The gradient of a scalar field represents?	Direction of a scalar field represent	Direction of maximum increase	Direction of zero change	Average rate of change
79	Which of the following is a vector quantity?	Energy	Power	Momentum	Work
80	A vector field with zero curl in a simply connected region is	solenoidal	Conservative	Rotational	Divergent
81	let f(x,y,z) = xyz. Find the gradiet of F at (1,1,1)?	i + j + k	2i + 2j + 2k	i + i + i	3i + 3j + 3k
82	A necessary condition for applying Stokes' Theorem is that the vector field must be:	Differentiable	Bounded	Conservative	Irrotational
83	Gauss Law states that the direction of E proportional to?	Electric flux	Charge density	Magnetic field	Currrent density
84	In the context of vector field E(x,y,x), the divergence of F at a point physically represents	the intensity of rotation of the field about that point	The magnitude of the vector field at that point	The net outward flux per unit volume	the total energy store
85	The value of auto correlation function at lag n = 0 is eqaul to th signals?	Average power	Total energy	RMS value	Peak Value
86	In a digital communication system, what does the correlation operation primarily determine?	the relation of frequency content between input nd channel response	The similarity measure between the received and reference signals	The comparison of average energy levels between transmitted and received signals	The effect of phase alignment in the reference and incoming signals
87	A ssytem is stable in the z doimain in the ROC?	Includes the origin	Incldues the unit circle	lies inside the unit circle	Lies outside the unit circle only
88	What is the result of circular convolution of two sequences x[n] and h[n]?	circular correlation	Linear convolution	Multiplication in the z-domain	Periodic extension of linear convolution
89	If the signal are orthogonal their cross-correlation is ?	Maximum	Minimum	Zero	Infinite

90	Two filters are cascaded. The first filter hasthe difference equation y1[n] = 0.5[n] + 0.5x[n - 1] and the sec filter has the difference equation y[n] = 0.2y1[n] + 0.8y1 [n - 1]. What is the difference equation of the overall cascaded system?	y[n] = 0.1x(n) + 0.5 x(n-1) + 0.4x(n-2)	y[n] = 0.25x[n] + 0.75x[n-1]	y[n] = 0.6x[n] + 0.4x[n-1]	y[n] = 0.1x[n] + 0.9x[n-1] + 0.4 x(n-2)
91	The inverse z-transform of X(z) = z/z-a correspond to?	$a^n u[n]$	δ[n]	u[n]	$(-a)^n$ u[n]
92	Given two discrete-time sequences x[n] = {1,2,0,1} and h[n] = {2,1,1,0}. Find the 4-point circular convolution y[n] = x[n] ⊛ h[n].	{4,5,3,4}	{3, 6, 3, 4}	{4,4,4,4}	{5,3,4,4}
93	Which statement best describe its response?	Underdapmed with small overshoot	Overdamped with no overshoot	Critically damped with fastest non-oscillatory response	Unstable with growing oscillations
94	For a unity feedback system with open loop transfer function G(s) = K/[s(s+2)]. What is the system type and the expected steady-state error for unit step input?	Type 0, non zero	Type 1, zero	Type 2, zero	Type 1, infinite
95	There are no loop. What is the overall transfer function from input to output?	2	6	4	8
96	The velocity error constant kv of a Type 1 unity feedback system is 10. What is the steady-state error for a unit ramp input?	0	0.1	10	Infinite
97		Poles are symmetric about the imajinary axis	All zeroes lie in the left-half plane	All poles lie in the left-half plane	The DC gain is less than 1
98	For the unity feedback system with open-loop transfer function G(s) = k/[s(s + 2)]. How many branches does the root locus have?	0	1	2	3
99		The average of the zeros only	The average of the poles only	(sum of poles - sum of zeros)/(number of poles-number of zeros)	(sum of zeros - sum of poles) -(number of poles + number of zeros)
100	A closed-loop transfer function is given by T(s) = 5/(s^2 + 6s + 5). What are the poles of the system?	-1 and -3	1 and 5	-1 and -5	0 and -5

GEAS	QUESTION	A	B	C	D
1	Which explanation Best account fr this observation?	Insufficient food and habitat support	Genetic mutation in the population	Improved environmental conditions	Increased reproductive efficiency
2	A sealed balloon is warmed gradually while external pressure remains unchanged. Which change is most likely be observed?	A two gas particles slow down	The Balloon expands outward	The gas density increases	The balloon's mass increases
3	From cost perspective these payments are considered?	Avoidable expenses	Variable operational cost	Marginal production costs	Fixed operating costs
4	A vibration travels along a stretched string while each segment at the string moves only vertically. What does this motion indicate?	The wave is longitudinal	Energy moves without matter displacement	The medium undergoes compression	Particle motion matches wave direction
5	What is the most logical step for the project manager?	Ignore minor cost overruns	Redefine the project vision	Close the project early	Analyze variance and adjust plans
6	During silicon wafer production, impurities cause inconsistent electrical behavior. Which stage is most critical to prevent this issue?	Crystal growth phase	Circuit etching step	Wafer polishing stage	Packaging process
7	How will the induced curret compare?	The slower conductor produces more	The faster conductor procudes more	Both produce equal current	Neither produces current
8	A small imperfection repeats throughout a crystal lattice, what is most likely the consequence?	Altered mechanical or electrocal behavior	No measurable effect	Increased molecular mass	Uniform chemical composition
9	What principle should take precedance?	Financial performance	Professional accountability	Obedience to management	Workplace harmony
10	A startup delays rapid expansion despite high demand. What is the most reasonable justification for this decision?	Limiting innovation	Avoiding competition	Preserving operational stabilty	Reducing customer interest
11	Which operational decision MOST directly improve customer satisfaction?	Reducing variation in service delivery duration	Adding multiple service options simultaneously	Increasing advertising frequency across platforms	Introducing advanced features ahead of demand
12	Which function is used C++ to create a window using the Windows API?	Initwindow()	setupWindow()	CreateWindow()	makeWindow()
13	Which provison requres ECE to support agriculture and Fish?	Section 27: Research and Development in Agricultural Technology	Section 34: Regulatory Standards for Equipment in Farming Practices	Section 15: Automation in Agricultural Processing	Section 21: Integration of Electronics in Agriculture and Fisheries
14	Which object-oriented programming concept allows method with the same name in different classes to exhibit different behaviors?	Inheritance	Encapsulation	Abstraction	Polymorphism
15	Which outcome is MOST likely under this managemet approach?	Greater predictability in organizational operations	Increased informal decision-making practices	Faster adaptation to environmental changes	Higher dependence on employee creativity
16	What is a bond?	Contract for property ownership	Document represents equity in a corporation	Written acknowledgement of debt	Certificate of Ownership
17	What is the most significant benefit of this approach?	Higher pricing flexibility after product launch	Improved alignment between product and user needs	Faster entry into competitive markets	Reduced dependence on promotional strategies
18	A circuit breaker trips repeatedly even when connected devices are minimal. After eliminating overload as a cause, what should be examined next?	Brand compatibility of connected electrical devices	Internal wiring integrity & possible insulation faults	Rated voltage of the electrical distribution panel	Installation date of the circuit breaker unit
19	This performace is BEST?	Productive but misaligned with objectives	Efficient but ineffective in goal achievement	Effective with poor resource utilization	Balanced in both efficiency and effectiveness
20	In CAD drafting practice, why are reference images commonly placed on a dedicated layer?	to automatically scale images with dimensions required	To allow selective visibility during drawing work	To improve printing resolution of drawings	To increase software rendering performance
21	What best explain this difference/	The amplitude of vibration	The wavelength produced by the source	The energy carried by the waves are different	The physical characteristics of the medium
22	What is the SI unit of electric current?	Watt	Ampere	Ohm	Volt
23	Which factor most influence its long term environmental viability?	End-of-life handling and recycling impact	Manufacturing cost of the battery	Availability of raw materials for production	Initial performance efficiency metrics
24	During SWOT analysis, which situation would most likely be classified as an external influence?	Investment strategy for product development	Employee training & skill enhancement	Changes in environmental compliance laws	Internal process optimization initiatives
25	Which keyword in C++ and Jave handle exception?	Exception	Error	Catch	Handle
26	A steel rod returns to its original length after being streched within its elastic limit. Which material property is demonstrated?	Toughness	Hardiness	Elasticity	Ductility
27	Assigning task deining responsibility and coordinating resources and activities most closely associated with what management?	Organizing	Leading	Planning	Controlling
28	Which requirement of the National Building Code most directly ensures public safety during earthquakes?	Paper documentation of permits	Adherence to structural and seismic design standards	Compliance with zoning regulations and protocols prepared ahead of time	Inspection of construction timeliness and correctness of documentation
29	In practical engineering systems, energy conservation MOST realistically mean?	Preventing energy conversion between forms	Maximizing useful output from available energy	Eliminating all forms of energy loss convert to useful energy	Generating unlimited energy supply
30	What battery type is commonly used by Uninterruptable Power Supply (UPS)?	Alkaline	Lead-acid	Lithium-Ion	Nickel-Cadmium
31	Which process most directly increase the acidity of ocean water over time?	Release of alkaline minerals from coral reefs	Dissolution of heavy metals from sediments	Mixing of fresh water and saltwater systems	Absorption of atmospheric carbon dioxide by sweater

32	In strategic marketing, which decision has the greatest impact on long-term competitive positioning?	designing short-term promotional campaigns	Adjusting product prices frequently	Selecting appropriate distribution channels	Determining tarket market segment
33	The theory X and theory Y which describe two different management tyles toward employee	Chester Barnard	Mary Parker Follett	Douglas McGregor	Elton Mayo
34	If current remains constant in a circuit and resistance increases, what happens to the voltage according to Ohm's Law?	Votlage becomes independent of resistance	Voltage remains unchanged	Voltage decreases proportionally	Voltage increase proportionally
35	Which characteristic of ceramic is most closely linked t their dominant bonding type?	Resistance to higher temperatures	Strong magnetic response	High electrical conductivity	Ability to deform plastically
36	A 10 Ω resistor connected to a 12 V battery. What is the current flowing through the resistor?	10A	12 A	0.83 A	20 A
37	In electronic device fabrication wha advantage do nanomaterial MOST directly provide?	Simplified manufacturing processes because of their size	Elimination of material waste	Lower demand for skilled labor because of greater production	Enhanced performance at reduced device size
38	Which topic would most likely be analyzed under macroeconomics?	National unemployment trends	Cost structure of a manufacturing plant	Consumer demand for smartphones	Pricing strategy of a single electronics firm
39	In which situation does particle motion occur parallel to wave propagation?	Ripples forming on a water surface	Light travelling through a vacuum	Compression waves moving through air	Electromagnetic radiation in space
40	Under RA. 9292, What is the primary reason unlicensed practice of electronics engineering is prohibited?	to increase licensing revenue	To ensure public safety & professional accountability	To limit competition within the profession	To protect employer interest
41	Which charge carrier becomes dominant in electrical distribution?	Holes created in the valence band	Free electrons in the conduction band	Protons bound within the lattice	Neutrally charged silicon atoms
42	During beta decay which particle is emitted from the nucleus as a result of the interaction involved?	gamma photon	Photon & Gamma waves	Electron or positron	Alpha Particle
43	Which factor most strongly support a funding decision?	Short-term revenue acceleration	Public visibility of the startup brand	Ability to pivot business models quickly	Evidence of scalable market demand
44	What type of bond hold carbon atoms together in graphite?	van der waals	Metallic	Covalent	Ionic
45	In series circuit containing resistor how electrical energy is distributed as current flows?	Stored uniformly in all components	Converted into heat across resistive elements.	Maintained at constant potential throughout	Transformed entirely into kinetic motion
46	Neutral paper fragment move towards a charged comb due to which electrostatic effect?	permament magnetization of paper	charge induction within paper	Uniform charge repulsion	Transfer of electron to the paper
47	Which condition is ESSENTIAL for sstaining nuclear fusion reaction in suns core?	Presence of heavy radioactive elements	Extremely high temperature and pressure	Constant supply of free neutrons	Low-density plasma conditions resulting to chain reaction
48	When white light passes through a prism, which property of light causes color separation?	reflection difference at the prism surface	Absorption of specific wavelengths by glass materials	Varation of speed with wavelegnth inside the medium	Interference between overlapping waves
49	Prolonged coral bleaching event MOST directly disrupt which ecological process?	Tidal energy distribution	Atmospheric heat transfer	Ocean current circulation	Marine food web stability
50	Elton Mayo's work highlighted the influence of social factors on productivity. What conclusion emerged from this research?	output depends solely on physical working conditions	worker behavior is shaped by social interaction	Authority structure determines employee motivation	Productivity increases only through financial incentives
51	Which section of PEC addresses electrical installation in hazardous location such as area of flammable gasses or vapors?	Regulations of equipment used in hazardous environments	Provisions on wiring methods and materials upon procurement	Requirements for grounding and bonding systems	General rules on electrical installation practices
52	In rigid body rotation, what characteristic remains unchanged for all particles of the body?	linear velocity magnitude	direction of tangential acceleration	Distance from the axis of rotation	Instantaneous linear momentum
53	At resonance in series AC circuit which condition causes the circuit to draw maximum current?	Capacitive reactance dominates the circuit	Inductive reactance exceeds resistance	Power factor becomes zero	Net reactance becomes zero
54	Under the Professional Electronics Engineering Law, who may legally supervise electronics engineering work performed by non-licensed personnel?	licensed technicians acting independently	Licensed Electronics Engineers with valid registration	Project managers appointed by the employer or company	Any registerd engineer
55	In an electromechanical reaction the anode is defined by whch fundamental process?	Gain of electrons by species	Loss of electrons by species	Neutralization of charged ions	Absorption of free radicals
56	Which factor MOST strongly limit the ability of low income countries to pursue long-term sustainable development?	insufficient access to clean energy sources	weak enforcement of environmental regulations	Dependance on imported technologies	Persistent poverty affecting education and health
57	What financial concept is used to determine its accumulated value at a future date?	Future worth of ordinary annuity	Simple interest computation	Capital recovery method	Present worth analysis
58	An engineer authors original instruction manual for commercial publication. Which form of protection best safeguards the expression of this work?	Trademark ownership	Trade secret classification	Patent protection	Copyright registration
59	Which programming approach emhasizes immutability and avoids changing program states?	Procedural programming	Imperative Programming	Object oriented programming	Functional programming

60	In object-oriented analysis, identifying real-world object attributes and interactions corresponds to which system aspect?	software architecture selection	user interface design	Problem domain modeling	Harware Platform Definition
61	Which responsibility of PRC most directly ensure continued competence among Electronics Engineer?	Accrediting engineering schools nationwide	Approving engineering project proposals	Enforcing ethical standards through disciplinary actions	Monitoring compliance with license renewal requirements
62	Which atmospheric process allows greenhouse gases to contribute to global warming?	reflection of solar radiation back into space	scattering of ultraviolet light by aerosols	Conversion of heat into kinetic wind energy	Absorption and Reemission of infrared radiation
63	A refrigerator absorbs 500 J of heat from the cold reservoir while 150 J of work is supplied to the system. What is the coefficient of performance?	1.5	2.5	3.33	0.3
64	Which stage of the water cycle contributes most to regulating atmospheric humidity over forested regions?	surface runoff into rivers	condensation in cloud formation	Transpiration from vegetation	Infiltration into groundwater
65	Which issue is most directly involve?	Improper contract negotiation	Violation of consumer protection laws	Breach of intellectual property rights	Failure to meet safety compliance
66	In Object-Oriented programming, why are objects considered fundamental building blocks of a system?	they control program execution flows	They store data independently of logic	They encapsulate data with related behavior	They define system architecture layers
67	In a supercapacitor effiecient energy storage relies heavily on which electrolytye function?	Rapid transport of ions during charge and discharge	Structural reinforcement of electrode material	Electrical insulation between electrodes	Thermal regulation of the device
68	Which feature distinguishes object-oriented programming from procedural programming?	dependence on compiler optimization	Because of variables and data types	Sequential execution of instructions	Grouping data with related methods
69	Which physical quantity captures tihs effect?	Instantaneous acceleration	Rate of change of energy	Work done on the object	Impulse delivered to the object
70	In a stationary fluid, why does pressure increase as depth increases?	molecular speed decrease with depth	Gravitational force acts on the fluid above	Fluid density becomes negligible	Fluid viscosity increases with depth
71	Which outcome BEST explain hy procedures are essential for energy flow in and ecosystem?	They convert solar energy into chemical energy.	They recycle nutrients back to the soil	They regulate population sizes directly	They consume organic matter for survival
72	A straight conductor moves perpendicular to a uniform magnetic field. Which change will increase the induced electromotive force?	decreasing the magnetic field strength	Increasing the speed of motion	aligning motion parallel to the field	Reducing the conductor length
73	What happens to the current?	It is reduced by half	It increases proportionally	It becomes zero	It remains unchanged
74	An engineer develops a novel electronic circuit with a unique functional design. Which protection most directly secures exclusive rights to this invention?	patent registrations	Copyright protection	Trademark registration	Trade Secret Classification
75	What is the SI unit of pressure?	Newton	Pascal	Joule	Watt
76	When light enters a transparent medium from vacuum, which factor directly determines the material refractive index?	change in wavelength relative to vacuum	Speed reduction of light within the medium	Angle of refraction at the boundary	Intensity loss during transmission
77	Which property most directly influences electrical conductivity in solid material?	Surface roughness	Thermal expansion coefficient	Availability of charge carriers	Crystal grain size
78	Which charactersitic distinguishes polymers commonly used in engineering applications?	strong metallic bonding between atoms	Ionic bonding with rigid lattice	High melting point and brittleness	Repeating long-chain molecular structure
79	In engineering economics the principle of equivalence allow engineering?	Predict market demand accurately	Eliminate interest rate effects	Determine tax obligations on investments	Compare monetary values at different times
80	Which crystal defect involves a missing atom at a lattice site?	Grain boundary	Vacancy defect	Interstitial defect	Edge Dislocation
81	Whch library BEST fits this requirement?	java.awt	javax.swing	jave.io	java.nio
82	A manufacturing firm reports record profits but faces community protests and evironmental fines. Which Triple Bottom Line dimension is most clearly being neglected?	Environmental Responsibility	Long-term innovation	Economic performance	Financial Sustainability
83	Which tool be used?	Radial dimension	Linear dimension	Angular dimension	Leader annotation
84	A motion analysis describes how an objects position charges over time, without mentioning mass or applied forces, which field of study is being used?	Kinematics	Statics	Kinetics	Dynamics
85	Which internal property most likely explains the difference?	Crystal color	Atomic mass	Surface texture	Electron mobility
86	A gas is slowly compressed while heat is allowed to escape so that its temperature does not change. Which condition BEST describes this process?	constant volume process	Isothermal process	Constant pressure process	Adiabatic Process
87	At higher frequency which change is observed?	Lower inductive reactance	Constant current magnitude	Zero phase difference	Higher inductive reactance
88	A programmer assigns an integer to a variable, then later assigns a string to the same variable without erros. Which language feature allows this?	static binding	Strong compilation rules	Manual memory allocation	Runtime type checking
89	Which principle is most evident in this decision?	Ethical responsibility	Cost minimization	Brand expansion	Market competitiveness

90	Before investing in mass production, a startup analyzes customer problems, purchasing behavior, and unmet needs. What stage of business development is this?	revenue optimization	Market validation	Product scaling	Operations Planning
91	Which ecological change MOST directly link these outcome?	Increased competition among remaining plant species	Faster growth cycles of invasive organism	Reduced primary productivity affecting ecosystem services	Improved adaptation of wildlife to urban conditions
92	An electronics engineer is offered consultancy work that conflicts with a current clients interests. According to Professional Regulations Commission (PRC) and RA 9292 guidelines, what action is most appropriate?	disclose the conflict and avoid unethical engagement	Proceed as long as no formal complaint exists	Delegate the work to a non-licensed assistant	Accept the work if compensation is higher
93	What is the most reasonable justification for this approach?	Reduction of operational complexity and increasing people interests	Elimination of competitive pressure	Enhancement of long term corporate sustainability	Compliance with marketing trends and people likeness
94	A plastic component softens when heated and regains shape upon cooling without chemical change. Which property does this behavior BEST illustrate?	High thermal degradation resistance	Permanent structural rigidity	Cross-linked molecular structure	Thermoplastic reprocessability
95	What advantage do these methods provide?	They permit interfaces to store object state.	They allow interfaces to maintain backward compatibility	They replace the need for abstract classes	They enable multiple inheritance of fields
96	A company redesigns how products are delivered to customers but keeps pricing and promotion unchanged.. which element is being adjusted that lies outside the traditional 4Ps?	Product	People	Process	Place
97	Which battery chaacteristic is MOST critical for this application?	Long shelf life without use	High tolerance to deep discharge	High energy density per unit mass	Low internal resistance
98	When analyzing multiple project options with uncertain outcomes, what is the primary advantage of using a decision tree?	avoids the use of probability estimates	Guarantees the highest possible profit	Eliminates the need for sensitivity analysis	Structures complex decisions into sequential choices
99	Which depreciation pattern does this describe?	Usage-based depreciation only	Residual-value-driven depriciation	Accelerated depreciation in initial periods	Uniform depreciation over time
100	Which crystal imperfection MOST directly alter charge carrier movement in semiconductor materials?	Lattice dislocations	Macroscopic shape irregularities	Grain boundary formation	External surface roughness

ESAT	QUESTION	A	B	C	D
1	... can correctly interpret the boundaries of the data frames in a communication system?	Frame synchronization	Network sytnchronization	Word synchronization	Bit-Level Synchronization
2	When setting up a communication system within the VHF and UHF frequency ranges, what factors must be considered for signal reliability in terms of the distance?	Global distance	shielded locations	Over the horizon distance	Line of sight distance only
3	FIR filter has differential equation y[n] = 0.3x[n] + 0.6x[n-1] + 0.1x[n-2]. What is the length of its impulse response?	3	Infinite	4	2
4	An AM transmitter operates with a carrier frequency of 1 MHz and is modulated by an audio signal with a maximum frequency of 5 kHz. What is the total bandwidth required for the transmitted AM signal?	2 MHz	1 MHz	5 kHz	10 kHz
5	For a given probability error binary constant FSK is inferior to binary coherent PSK by approximately?	2 dB	6 dB	0 db	3 dB
6	A receiver has a noise figure of 6 dB. By what factor is the noise increased at the receiver?	2	10	8	4
7	... What is the transmission rate on the TPM link if the frame efficiency is 90%?	8.44 Mbps	12.44 Mbps	4.44 Mbps	16.44 Mbps
8	A rectangular waveguide operates above cutoff. Two signals at 14 GHz and 18 GHz are transmitted through the core waveguide section. Which statement is correct?	The higher frequency signal arrives later	both signals arrive at the same time	The lower frequency signal arrives later	Arrival time depends only on waveguide length
9	helical antenna is designed for 1.5 GHz operation in axial mode. What is the optimum spacing per turn?	25 mm	50 mm	75 mm	100 mm
10	When is a signal considered even?	f(t) = - f - (t)	f(t) = f(−t)	f(t) = f(t + T)	f(t) = -f(t + T)
11	source generates 6 different messages with probabilites: 1/4, 1/4, 1/8, 1/8, 1/8, and 1/8. What is overall entropy of the source?	0.25 bits	0.50 bits	2.50 bits	5.00 bits
12	What is maximum data transfer rate provided by the Basic Rate Interface (BRI) of ISDN?	128Kbps	1.5 Mbps	4 Mbps	64kbps
13	What is the differential equation of the entire cascaded system?	y[n]= 0.1x[n] + 0.5x[n-1] + 0.4x [n-2]	y[n] = 0.25x[n] + 0.75x[n-1]	y[n] = 0.6 x[n] + 0.4 x[n-1]	y[n]= 0.1x[n] + 0.9x[n-1] + 0.4x [n-2]
14	Which statement about ERP and EIRP is correct?	ERP is referenced to an isotropic radiator	EIRP is always lower than ERP	ERP is referenced to a half-wave dipole	ERP and EIRP are always equal
15	Two signals, 17 GHz and 12 GHz, are transmitted through a 50m long section of this waveguide. What is the difference in their arrival times at the end of waveguide?	65.5 ns	105.4 ns	95.5 ns	56.6 ns
16	A 450 Watt carrier is simulatneously and sinusoidally modulated by three audio waves. The corresponding modulation percentages are 85, 42.5, and 21.5, respectively. If the resultant AM wave is propagated through 50 ohms antenna, the total radiated current is?	4.42 Amp	3.64 Amp	3 Amp	663.36 Amp
17	Calculate the amplifier's noise factor F at a room temperature T = 300 K.	0.34 dB	1.24 dB	0.67 db	1.16 dB
18	A PAM signal is generated using a natural sampling technique. The highest frequency component of the analog signal is fm. What is the minimum sampling frequency to avoid aliasing in the PAM signal?	fm	4 fm	fm/2	2fm
19	Critical angle if propagation in the D-layer, approximately 70 km in height, if the transmitter and receiver are separated by 500 km?	7.97°	9.13°	18.25°	15.64°
20	What is the purpose of a bridge in networking?	To connect two different networks and route between them	to ensure secure data transmission between devices	To divide a network into segments and filter traffic between them	To assign IP addresses to devices
21	In a smith chart, what does the center of the chart represent?	Infinite resistance	Matched impedance	An open circuit	A short circuit
22	Why is a four-wire circuit generally prefered for high-speed communication systems?	It eliminates the need for complex encoding schemes during transmission	It allows simultaneous transmission and reception of data, improving throughput and reducing latency.	It uses a single wire pair, reducing physical insfrastructure requirements	It requires less power for long distance signal propagation
23	What is the effect of the AGC circuit that has a time constant that is too short?	The gain of the receiver will fluctuate rapidly	The feedback loop will react slowly to every input variation	The receiver will maintain a constant output volume	The receiver will not be susceptible to any interference.
24	A digital filter has a transfer function H(z) with a pole at z = 1. What will be the stability condition of the filter?	The fiter is stable because the poles lies in the real axis	The filter is unstable because the pole lies outside the unit circle	The filter is marginally stable because the pole lies on the unit circle	The filter is always stable regardless of pole location

25	Which of the following best describes FIR filter?	It is less stable than IIR filters	It has finite number of nonzero impulse response coefficients	It requires the use of poles for analysis	It always has feedback
26	An engineer implemented cookies to manage user sessions. Which statement is correct?	it includes an expiry date and time in the hyperlink	It is delivered to the user's using an HTTP header.	It is a small text file stored in a user's browser	It is used to track a user's browsing pattern
27	, defined as the ratio of the output current to the optical power. Which factor indicates conversion efficiency?	Dark Current	Responsivity	Transit time	Acceptance Cone
28	In a piezoelectric transducer, electric charge is generated due to?	vibration of the crystal under mechanical stress	variation of temperature	Magnetic field changes	Electromagnetic radiation
29	Which multiplexing technique is MOST suitable for real time data such as video conferencing?	TDM (Time Division Multiplexing)	CDM (Code Division Multiplexing)	FDM (Frequency Division Multiplexing)	WDM (Wavelength Division Multiplexing)
30	Which of the following digital modulation schemes is most commonly used in wireless communications system due to thier robustness in noisy environments?	Binary Phase Shift Keying (BPSK)	Quadrature Amplitude Modulation (QAM)	Frequency Shift Keying (FSK)	Phase Shift Keying (PSK)
31	, which of the following is the impact on the spectral efficiency of the system?	The system's spectral efficiency is unaffected regardless of the modulation index	The system requires more power to maintain the same spectrial efficiency	Spectral efficiency decreases because higher modulation depth increases bandwidth	Spectral Efficiency improves due to the increased modulation depth
32	Which of the following emerging technologies is commonly integrated into modern building management (BMS) to improve energy efficiently?	3D printing	Internet of Things (IoT)	Artificial Intelligence (AI) for facial recognition	Blockchain
33	SSB transmission, why is the carrier suppresed?	To maximize the signal-to-noise ratio at the receiver	To reduce power consumption without affecting the transmitted information	To simplify the receiver design	To minimize the bandwidth required for transmission
34	You are analyzing the programming options for a programmable logic controller. Considering the needs of industrial automation systems, which of the following programming languages would be the most suitable for programming PLCs?	Ladder Logic	Python	SQL	Java
35	What does CRC do in error detection?	It uses a generator polynomial to divide the message and checks the remainder	It corrects error by retransmitting the data	It encodes data to minimize transmission errors	It detects errors in transmission by ading parity bits
36	Which of the following processes would occur in the the network layer when an IP packet is sent from one device to another?	Reordering out of sequence packets	Compression of packet headers	Determine the route the packet should take to its destination	Error correction of corrupted packets
37	In CDMA, what happens when multiple users transmit on the same frequency at the same time?	The system automatically shifts users to different frequencies	The signals are separated using unique codes	The system detects and corrects the interference	Their signals are separated by using different time slots
38	Which of the following conditions ensures a signal is sampled without alising?	The sampling frequency is more than twice the highest frequency component	The sampling frequency equals the Nyquist rate.	The sampling frequency is less than twice the highest frequency component	The signal is band limited, but the sampling frequency is arbitrary
39	degree-out-of-phase carrier signals. What will be the outcome when you increase the number of constellation points?	Bit error rate remains the same	Data rate increases	Data rate decreases	Bit error rate decreases
40	What action would be the most appropriate response if a fire alarm system in a building management system detects a potential fire hazard in a building?	Increase temperature for regulation	Alert occupants and start evacuating	Improve air circulation to reduce smoke	Enhance lighting for visibility
41	... for combining multiple data streams into a single one for more efficient transmission over a communication system?	Terminal	LCU (Line Control Unit)	FEP (Front end processor)	Multiplexer
42	What happens to the group velocity of a waveguide mode as the operating frequency approaches the cutoff frequency?	It approaches the speed of light	It becomes infinite.	It approaches zero	It remains constant
43	modulation scheme used for satellite communication due to its constant envelope?	GMSK	ASK	PSK	QAM
44	In PAM, the signal is sampled, and each sample is quantized to aspecific amplitude. Which of the following represent the main limitation of PAM in terms of signal distortion?	Signal distortion due to the loss of high frequency components in the signal	Increased noise susceptibility due to quantization error.	Reduced bandwidth efficiency due to high number of amplitude level	Intersymbol Interference (ISI) resulting from insufficient sampling rate
45	How would a company with offices in multiple cities benefit from using WAN compared to LAN?	WAN uses fiber optics, while LAN uses copper cables exclusively	WAN is ideal for data transfer between devices in a single office	WAN connects multiple geographically separated locations over long distances	WAN allows for higher-speed, local data transfers within the same building
46	Two amplifiers are compared at 300 K, Amplifier A: Te = 60 K, Amplifier B: Te = 120 K, which statement is correct?	Amplifier A has higher noise factor	Amplifier B has lower noise figure.	Amplifier A has lower noise figure	Both have equal noise factor
47	... in digital communication system, what is typically used to represent information in a form that can be transmitted over the channel.	Electromagnetic waves	Analog signals	Continuous waves	Discrete sybol or bits

48	What will be the effect of noise on the performance of a digital communication system when signal to noise ratio is significantly reduced?	It does not affect the performance at all	It reduces the bandwidth.	It increases the bit error rate	It decreases the data rate
49	, what happens to the received power density if the distance is doubled?	It remains the same	Iti s reduced to half	It is reduced to one-fourth	It is reduced to one-sixteenth
50	A programmable logic circuit needs to control a high-power DC motor. Which interfacing methods is most appriorate to use?	Using a relay switch in the motor's power	Direct connection to the PLC's output module	Using a Digital-To-Analog Converter (DAC) to control the motor speed	Connecting the motor directly to the AC mains
51	Modulation technique most suitable for high-speed data transmission in modern wireless and cable communication system?	ASK	FSK	QAM	DPSK
52	Which measure primarily protects a BMS network from external cyber threats and malicious traffic?	Physical security looks	Regular HVAC maintenance	Network firewalls	Manual system overrides
53	Which parameter indicates perfect impedance matching in a transmission line system?	High VSWR	Infinite return loss	VSWR equal 1	Large Standing wave ratio
54	A broadcast station upgrades its antenna system to reduce reflected power and improve efficiency. Which adjustment directly addresses this objective?	Increasing transmitter output power	Shortening the transmission line	Matching the antenna impedance to the transmission line	Operating at a lower frequency
55	Which component of the radio receiver determines the tuning of the desired station?	Audio amplifier	Detector	Local Oscillator	Power supply
56	A tranmission is terminated with a load impedance smaller than its characteristic impedance. What is the phase of the reflected voltage wave?	In phase with the incident wave	180° out of phase w/ the incident wave.	Leading the incident wave by 90°	Independent of impedance
57	If T = -0.3, how much power is delivered to the load?	7 W	8.2 W	9.1 W	10 W
58	If a communication problem involves signal attenuation or noise due to medium, which OSI layer is most likely affected?	Transport Layer	Network layer	Data link Layer	Physical Layer
59	At which OSI layer does this problem occur?	Data Link Layer	Network Layer	Transport layer	Physical Layer
60	Which characteristic is most associated with a peer-to-peer network?	Centralized control and management	High security & scalability	Dedicated server infrastructure	Decentralized resource sharing
61	In which network model do all participating devices have equal roles and can act as both client and server?	Client-Server	Mainframe	Cloud Computing	peer-to-peer
62	Parseval's theorem establishes an equivalence between which two representations of a signal?	Time Domain and Laplace Domain	Time domain & z-domain	Frequency domain and Z-domain	Time Domain and Frequency Domain
63	In amplitude modulation, what condition defines overmodulation?	Modulation index equal to 0	Modulation index less than 1	Modulation index equal to 1	Modulation index greater than 1
64	In free-space propagation, how does the Electric field strength vary with distance from the transmitter?	Directly proportional to distance	Proportional to the square of distance	Inversely proportional to distance	Inversely proportional to the square of distance
65	Which block in a digital communication transmitter performs frequency translation?	Source encoder	Channel Decoder	Digital Modulator	Demodulator
66	Which fiber type offers better bandwidth than step-index multimode fiber but is simpler than single-mode fiber?	Plastic Optical Fiber	Step-index single-mode fiber	Graded-index multimode fiber	Hollow-core fiber
67	Which statement BEST describes the refractive index profile of graded-index multimode fiber?	Constant throughout the core	Abruptly changes at the core-cladding boundary	Higher in the cladding than in the core	Maximum at the center and decreases toward the cladding
68	Which VPN feature allows corporate traffic to pass through the secure tunnel while permitting internet traffic to access the public network directly?	Full Tunneling	Encryption-only tunneling	Site-to-site VPN	Split Tunneling
69	IBM SNA, which component directly interface with end-user applications?	Physical Unit (PU)	Transmission Control Unit (TCU)	Data Link Control	Logical Unit (LU)
70	The direction of the Poynting vector indicates:	Direction of electric current	Direction of magnetic field oscillation	Direction of wave attenuation	Direction of electromagnetic energy propagation
71	optical fiber spcifically designed to support only one propagation mode, thereby minimizing model dispersion?	Step-index multimode fiber	Singlemode Optical Fiber	Graded-Index Multimode Fiber	Plastic optical fiber
72	A telecommunications company is deploying a multiple-access system where each user is assigned a unique frequency band to prevent interference. Which access tech is being used?	Assign a separate frequency band to user	Allows multiple users to share the same frequency simultaneously	Uses spreading codes for multiple access	Assigns unique time slots for users
73	primary advantage of digital communication over analog communication.	Digital signals are less affected by noise and interference	Analog signals offer better security than digital signals	Digital systems require more bandwidth than analog systems	Digital communication is always faster than analog communications
74	Thermocouples are used in temperature sensing applications. The voltage generated by a thermocouple is proportional to the:	Applied magnetic field	Difference in temperature between the two junctions.	Current passing through the thermocouple	Resistance of the metal used

75	Which characteristic of bus topology makes it unsuitable for large scale networks?	High installation cost	Limited transmission speed of devices	Poor scalability and performance under heavy traffic	Requirement of central hub
76	Which network topology experiences the MOST difficulty in troubleshooting due to shared transmission paths?	Star Topology	Ring Topology	Mesh Topology	Bus Topology
77	Which topology does the failure of a single node MOST likely affect only that node and not the entire network?	Bus Topology	Ring Topology	Star topology	Linear Topology
78	Which topology is most suitable for networks requiring easy fault isolation and centralized management?	Bus Topology	Ring Topology	Mesh Topology	Star Topology
79	Topology that provides the highest reliability through multiple redundant paths between nodes?	Bus Topology	Ring Topology	Star topology	Mesh Topology
80	If two communicating systems use different data representation, which OSI layer resolves this incompatibility?	Transport	Network	Data Link	Presentation
81	PRIMARY advantage of synchronous transmission over asynchronous transmission?	Simpler Hardware Implementation	Reduced need for timing information	Higher efficiency for continuous data transfer	Better error detection capability
82	In which application is synchronous transmission most suitable?	Keyboard input to a computer	Serial communication in microcontrollers	Low-speed sensor data transmission	High-speed data links and network backbones
83	Which relationship explains why broadband systems can support higher data rates?	Ohm's Law	Faraday's Law	Shannon- Hartley Theorem	Kirchoff's Law
84	According to the Nyquist Sampling Theorem, what is the MINIMUM sampling frequency required to avoid aliasing?	Equal to the signal bandwidth	Equal to the highest frequency component	Twice the highest frequency component	Four times the highest frequency component
85	Which condition leads to aliasing in a sampled signal?	Oversampling the signal	Sampling exactly at Nyquist Rate	Filtering before sampling	Sampling below the Nyquist rate
86	A new system was implemented in a UNIX environment. The server process must call several times in the correct sequence to establish a connection and then receive data. How will the order of calls, 2-bind, 3-listen, 1-accept, and 4-receive, be passive if there is a change in sequence?	It will be possible but only few functionalities will work	It is possible and will only work based on the identified sequence of function.	It will work properly even if the sequence changes frequency	It will not be possible as it prevents the server from functioning correctly
87	... for a more reliable bit detection in a communication channel. What is the relationship between SNR and BER?	BER increases as SNR increases	BER decreases as SNR increases	BER remains constant regardless of SNR	BER is independent of SNR and depends only on modulation type
88	Which satellite orbit is most suitable for continous communication coverage over a specific region?	Polar Orbit	Inclined orbit	Elliptical orbit	Geostationary Equatorial Orbit
89	In PAM, which parameter of the pulse is varied according to the message signal?	Pulse position	Pulse width	Pulse Amplitude	Pulse repetition rate
90	Compared to PAM, which pulse modulation techniques is more resistant to amplitude noise?	Pulse-Amplitude Modulation	Pulse-width Modulation	Analog Modulation	Envelope Modulation
91	Which of the following orbits is commonly used for geosynchronous communicating satellites?	Retrograde orbit	Equitorial Orbit	Inclined orbit	Polar bit
92	Which statement best describes a deterministic signal?	Its future values cannot be predicted	It varies randomly with time	It is characterized only by probability density functions	Its values are completely predictable for all time
93	Which property is NOT associated with deterministic signals?	Predictability	Exact mathematical representation	Statistical description using probability theory	Known future behavior
94	Compared to BPSK, QPSK primarily improves:	Noise immunity	Receiver simplicity	Bandwidth efficiency	Signal Amplitude
95	PSK systems, increasing the number of phase states generally result in:	Lower data rate	Reduced bandwidtth efficiency	Higher susceptibility to noise	Improved noise immunity
96	Which modulation technique is most suitable for long-distance HF radio communication?	Conventional AM	Frequency Modulation	Single Sideband (SSB)	Pulse Amplitude Modulation
97	Major disadvantage of SSB compared to AM is that it requires:	Higher transmitted power of the signal	Wider bandwidth	More complex transmitter and receiver circuits	Lower signal-to-noise ratio
98	The resonant frequency of a dipole antenna is most directly determined by its:	Radioation Resistance	Input impedance	Physical length relative to wavelength	Polarization
99	Impedance matching of an antenna primarily affects which parameter.	Resonant frequency	Radiation pattern	Power transfer efficiency	Physical Length
100	you are designing a signal processing system to filter a signal using a known filter impulse response. To calculate the output of the system, which mathematical operation would you apply between the input x(t) and the filter response y(t)?	$\int x(T)\cdot y(T)\,dT$	$\int x(t-\tau)y(\tau)d\tau$	$\int x(t)\cdot y(t-\tau)\,dt$	$\int x(t)\cdot y(t)\,dt$